Parametric Pavillon

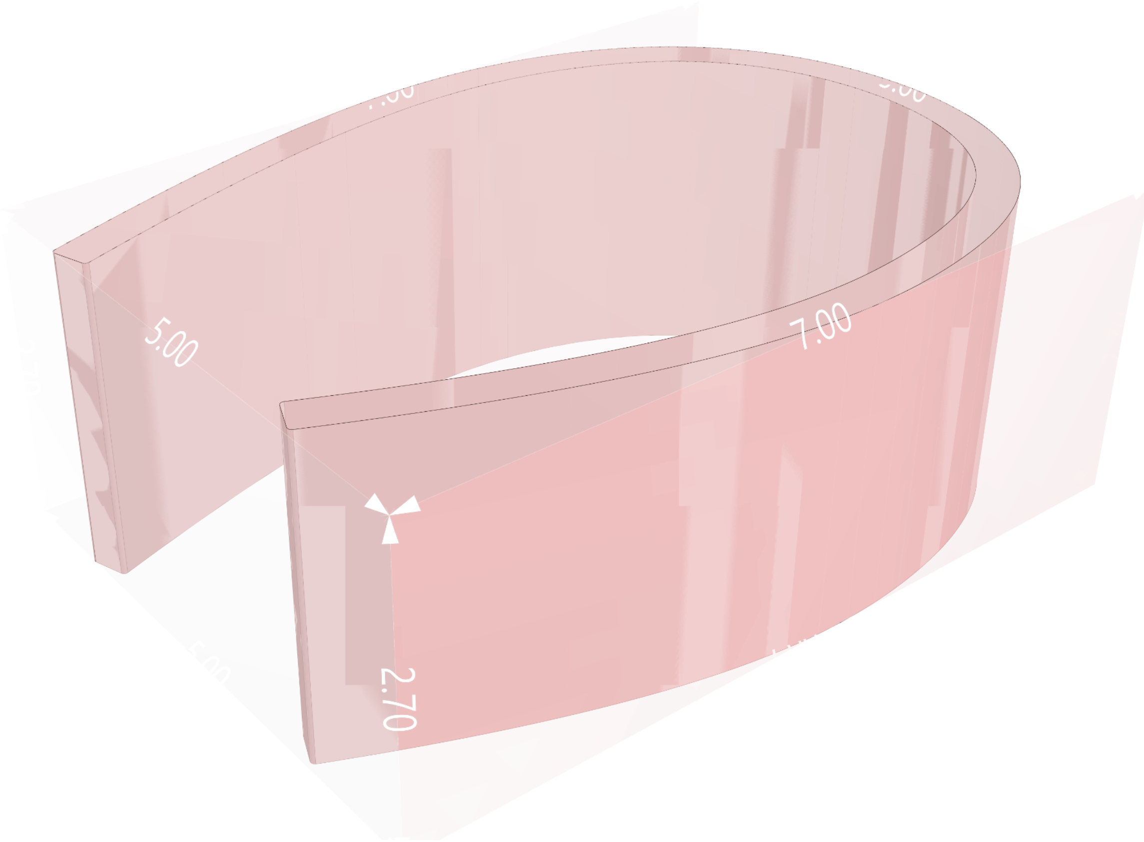
„facade mock up - house design - calculation“

Task: Design your own facade mock up, scale it to an exibition pavillon and calculate costs, masses and count the ankers.

Facade Mock Up: The piece of facade has to have a dimension of 2.33m lenght, min. 26cm, max, 45cm in width (shell distance) and 2.70m height. All outer edges has to be rounded by 3cm radius. The overhang shouldn`t be bigger than 8 degrees. Be careful with this limits, your print depends on this parameters. Your model has to be absolutely clean and shouldn`t have any gliches.

Pavillon: Your previous task has to be a work sample for the actually project. For this you have to translate your facade design to a pavillon curvature. The maximum dimensions of your pavillon are max. 7.00m in lenght, 5.00m in width and 2.70m in height. Your shell distances should be the same as in the first task (min. 26cm, max. 45 cm). Try something unique and organic, but be awared of the overhang limits, structural requirements and radii. Finally put an additional roof on it. But this will not be printed. So try to hold this simple. Invest work in the printed structure.

Calculation: For sure it is important to present work samples and first designs, but moreover for a client who pays money for a project it is the most important to calculate some things. Your last task is to calculate volumes (m³), masses (to) and costs (€/$).



Pavillon requirements

Dimensions wall: centerline: 2.33m lenght, 2.70m height, 26-45cm shell distance

Dimensions pavillon: 7.00m lenght, 4.00m widht, -2.70m height, 26-45cm shell distance

General design requirements

Overhang: -8° (degree)

Radius of edges: not under 3cm

Output: wall and pavillon as seperate STEP-Files with projectname (for example: wavybubbles_wall.step, wavybubbles_pavillon.step)

Have fun and we are very excited to print your work! See you tomorrow.